

Word Priming Experiment
PSY310 Lab in Psychology

Lab Report

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GitHub Link :

Introduction

A word priming experiment in PsychoPy is a psychological research, which implements the PsychoPy software, to present pairs of words sequentially to the participants and how the first word (the prime) affects the second word (the target). The process measures the response times or accuracy to interpret cognitive processes such as semantic association or word recognition. In this type of experiment, the participants will be presented with a prime word, after which a target word will be presented to the participants and the target will be expected to carry out a task on the target- to either determine whether a word is a real word, to name or to categorize it. The point is that when the prime is processed, the related concepts are activated in the memory, and then related targets are easier or faster to recognize.

The popular use of word priming experiments is to determine how previous exposure to one stimulus affects cognitive processing of the next stimulus. PsychoPy enables the researcher to manipulate the timing, randomisation, and design of such experiments with great accuracy. Experiments in word priming are popular in psychology, linguistics and neuroscience in order to research, for example, language processing, implicit bias research and a multitude of others.

Method

Participants

The experiment was performed by four participants. Participant 1 and Participant 2 were 18 year old undergraduate female students. Participant 3 and Participant 4 were 20 year old male and 20 year old female; both students at Ahmedabad University.

Materials & Procedure

The experiment was created on a laptop, using the PsychoPy v2025.1.1. 15 trials were conducted for Routine 1 and 20 for Routine 2 (Test) per participant.

The experiment had two routines. The first routine required three components - a text, a slider and a loop. The second routine consisted of - two texts (text_2 and text_3), a text box, a mouse response and a loop.

The participants in this task were initially presented with 15 words whereby they were required to rate their familiarity with each word. These are the words that constituted the study list, where they primed their responses on the next stage. Once the familiarity ratings were completed, participants were then asked to do a word-fragmentation task

which involved the 15 primed words (and 5 new (non-primed) words). In case of fragments, the participants were to type the whole word in 10 seconds.

A hit was a situation where the people wrote the same word that was in the study list (Figure 1) as the one they wrote in a familiarity rating test. An example is that it could be spelled both as choke and as phone; the familiarity rating test was done with the word *phone*. The participant was successful when he/she typed "phone." Any other key word typed was considered to be inaccurate. On the same note, it was considered a hit when the reaction to the nonprimed words was of the expected response. The next item was presented in case of time-exhaustion. To determine the proportional hit by the study list, the number of hits according to the study list words was divided by the total number of study list words. The proportional hit of the nonprimed words was calculated by dividing the amount of hits on the nonprimed words by the total amount of the nonprimed words. The priming effect was determined by subtracting the proportional hit of words that were not primed with the proportional hit of words of the study list.

	A	
2	book	
3	cushion	
4	phone	
5	marble	
6	board	
7	Laptop	
8	Chart	
9	T-shirt	
10	mouse	
11	Pastry	
12	Case	
13	Cruise	
14	Police	
15	Keys	
16	Poster	

Figure 1

	A	B	C
1	test_word	Reference	old_new
2	l_pt_p	Laptop	old
3	_ha_t	Chart	old
4	T_hi_t	Tshirt	old
5	_ou_se	mouse	old
6	P_as_tr_	Pastry	old
7	c_s_	Case	old
8	_ru_is_	Cruise	old
9	P_li_e	Police	old
10	K_y_	Keys	old
11	P_st_r	Poster	old
12	_o_rd	Board	old
13	_a_bl_	Marble	old
14	_ho_e	Phone	old
15	_oo_	Book	old
16	c_s_io_	Cushion	old
17	ta_l_	Table	new
18	p_nc_l	Pencil	new
19	v_s_	Vase	new
20	bo_t_e	Bottle	new
21	c_m_ra	Camera	new

Figure 2 : The list of test words that are used as reference.

After this, we run the experiment and the data will be automatically saved in an excel file.

Results

Participant 1

Priming score = -0.47

This has a negative priming effect, which makes the participant provide more false answer to the research questions. It is also possible that the participant subconsciously repressed the word when it was first presented and was unable to access it when the fragment of word was presented.

Participant 2

Priming score = -0.28

This has a negative priming effect, which makes the participant provide more false answer to the research questions. It is also possible that the participant subconsciously repressed the word when it was first presented and was unable to access it when the fragment of word was presented.

Participant 3

Priming score = -0.293

This has a negative priming effect, which makes the participant provide more false answer to the research questions. It is also possible that the participant subconsciously repressed the word.

when it was first presented and was unable to access it when the fragment of word was presented.

Participant 4

Priming score = -0.13

This has a negative priming effect, which makes the participant provide more false answer to the research questions. It is also possible that the participant subconsciously repressed the word.

when it was first presented and was unable to access it when the fragment of word was presented.

Discussions

The existing priming outcomes indicate that there was a negative priming effect, the subjects were less successful at reciting the items in the study list on average than they were at recognizing nonprimed items in the fragmentation task (mean priming score = $[-0.96]$). Such a significant negative priming implies that the exposure of these stimuli in the past did not result in an enhanced subsequent identification; rather, the exposure of these stimuli suggests that the previous exposure seems to have interfered or suppressed the retrieval of the stored representation during the test phase.

Similar negative-priming effects in a face-based form of such paradigm might occur in either of the following ways: (1) initial exposure to a face might lead to encoding of face using non-diagnostic features (e.g., clothing or background) that are deleted in the fragmented version, or (2) repeated exposure to a face might lead to reduced

dependence on the internal face template required to identify degraded faces. On the other hand, positive priming (positive increase in the identification of primed faces) would be expected in the case the task during the study-phase stimulates deep person-level encoding.

References

1. Kantowitz, B. H., Roediger, H. L., & Elmes, D. G. (2009). *Experimental psychology*. Boston, MA, USA: Wadsworth Cengage Learning.